

Multimodal Input Layer

 **Network Traffic
Features (CNN)**

 **System Logs
Sequences (LSTM)**

 **API Traces
Patterns (GRU)**

Specialized Encoding

CNN: 32 → 64 → 128

BiLSTM: 128 units

GRU: 64 units

 **Multimodal
Fusion**

Concatenation + LayerNorm
 $z_{\text{fused}} \in \mathbb{R}^{3dz}$

 **Stochastic
Transformer**

**4 Encoder Layers
8 Attention Heads
Gaussian Noise: $\varepsilon \sim N(0, \sigma^2 I)$**

Uncertainty Quantification

**Sparse Gaussian Process
RBF Kernel
Variational Inference**

Uncertainty Estimates
 μ^*, σ^{*2}

 **Adversarial Training**

**Multi-Attack Generation
FGSM | PGD | C&W
DeepFool | GAN**

 **Classification
Layer**

Uncertainty-Weighted Classifier
 $p(y|X, \sigma^{*2})$

 **Final Output**

**Attack Classification
+ Confidence Bounds
ECE = 0.031**